saleem khan

NRPU (8)

Document Details

Submission ID
trn:oid:::3618:156569062

Submission Date
Oct 30, 2026, 5:45 PM GMT+0

Download Date
Oct 30, 2026, 5:55 PM GMT+0

File Name
NRPU (8).pdf

File Size
78.7 KB

7 Pages

3,703 Words

21,863 Characters

1% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **3** Not Cited or Quoted 1%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

1% Internet sources
1% Publications
0% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- **3** Not Cited or Quoted 1%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

1% Internet sources
1% Publications
0% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1 Internet

vbn.aau.dk **<1%**

2 Publication

Ellsworth, Nathan A.. "From Observability to Orchestration: A Cloud-Native Mana... **<1%**

3 Internet

faculty.stevens.edu **<1%**

Selective Escalation from Rules to LLM Reasoning for Resilient 6G Network Orchestration

Title of the research	Selective Escalation from Rules to LLM Reasoning for Resilient 6G Network Orchestration
Theme / research domain	AI and Cybersecurity
Discipline	Defence and Strategic Technologies
Sub discipline and suggested reviewer domain	Agentic AI and multiagent systems for autonomous network management, and AI enabled 6G and O-RAN network orchestration. Suggested reviewers: experts in wireless network management with AI, in O-RAN and 6G RAN intelligent controllers, and in agentic systems built on LLMs.
Total duration	24 months, in two phases of 12 months
Industrial collaborator	To be completed by the PI.
International collaborator	To be completed by the PI.

Problem Statement

Sixth-generation (6G) network requires to coordinate different decisions like network-slice allocation, admission control, handover, and energy-aware operation under strict latency and reliability. To make such decisions, there are established control policies which handle these tasks in efficient manner in routine conditions. However, in disaster situation or or mission-critical operations, network conditions change abruptly. Some links may become unreliable, local failures can propagate, and service priorities may conflict. In such scenario, when the normal policies do not fit in the situation, there is need of network-management approach which should be capable to respond intelligently.

Existing network management uses the policy of rule-based and reinforcement-learning (RL). Although these solutions offer low latency and relatively predictable behavior, performance is degraded under the previously mentioned scenarios. Their performance deteriorates when decisions involve unfamiliar combinations of failures, competing objectives, or priorities that are absent from their design or training.

Large language models (LLMs), used within agentic AI workflows, may help interpret such ambiguous situations and coordinate responses across tasks. Applying an LLM to every control decision is nevertheless difficult to justify: inference adds delay and computational expense, and repeated remote calls may be incompatible with stringent service requirements. For sovereign, emergency-response, and defence networks, dependence on cloud-hosted reasoning services also raises questions about operational autonomy, connectivity, and control over sensitive data.

To cope with such scenarios, researchers proposed Large language models (LLMs), used within agentic AI workflows. These solutions help to interpret such complex situations to solve ambiguous situations with coordinated responses across tasks. However, escalating an every call to LLM to make decision is not efficient in terms of delays and computational expenses. Moreover, control over sensitive data, emergency-response, and defence networks, dependence on cloud-hosted reasoning services also raises questions about operational autonomy and connectivity.

Cost-aware LLM routing provides a useful starting point. Approaches such as RouteLLM, FrugalGPT, and Hybrid LLM select among models according to estimated query difficulty or expected quality, with published studies reporting substantial reductions in inference cost under their respective evaluation settings. However, model-to-model routing is not equivalent to deciding whether a reasoning model is

needed at all. Network orchestration includes decisions that can be executed safely through deterministic policies, as well as cases where contextual interpretation may be valuable. Traditional approaches mentioned previously use routing method with generic purpose and not designed to be able to make decisions based on the distinction to different scenarios.

Recent research on using agent-based AI solutions for 6G is expanding. These solutions include zero-touch managment, intent-based networking, and multi-agent control of radio-access and network resources. These studies indicate that coordinated agents can be useful for complex management tasks. However, they also expose a practical design question that for each decision how much reasoning is necessary.

In many architectures, the agentic workflow is invoked for every request. Where inference efficiency is considered, the emphasis is commonly on choosing between smaller and larger models or between edge and cloud execution. The decision to retain a routine task on a conventional rule path, and the impact of that choice on network-level outcomes, remains insufficiently evaluated.

Although,there are studies closest to this proposal exist in literature, however, it address only part of the problem. An agentic intent-based networking approach routes decisions through a multi-agent pipeline without first assessing whether the case warrants that level of reasoning. A network-aware routing approach adapts model selection to network conditions, but it does not include a deterministic rule-based option and evaluates general tasks rather than operational network-control decisions. Taken together, these limitations leave open the question of how to allocate reasoning selectively while preserving the reliability and timing expected of network management.

The research gap is therefore the absence of an integrated and systematically evaluated framework that (i) executes routine decisions through a rule-based pathway, (ii) uses an explicit escalation criterion to determine when LLM reasoning is justified, and (iii) evaluates this choice on representative 6G orchestration tasks under disaster and mission-critical conditions.

In this research proposal, we are going to fill the above mentioned gap by proposing a selective escalation based Hybrid cost-aware routing scheme. The idea is based on our published peer-reviewed hybrid multi-agent architecture which we are going to adapt for 6G network management in this work. Our previous work provides a starting point for this project. This study will take the idea further by developing an escalation strategy for 6G network management. It will be evaluated across network scenarios and in a standards-aligned environment. Proposed work will be assessed in terms of task success, decision latency and inference cost. These metrics must be assessed together because improving one metric at the expense of the others may not produce a control system.

Objectives and Scope

Following are the project objectives of our proposed work. These objectives are build on a previously evaluated hybrid multi-agent architecture which we are going to extend for different tasks specific to 6G network management.

1. **Define an escalation method for 6G network decisions.** We will develop decision-complexity indicators for five 6G management tasks. These tasks include admission control, slice allocation, handover, self-healing, and energy management. Based on the estimated complexity and uncertainty, an escalation policy will assign routine decisions to rule-based controllers while invoke LLM agents only when more advanced reasoning is required. This approach will balance decision quality, efficiency, and computational cost.

2. **Extending published hybrid architecture into network management.**In our previous work, we proposed the architecture that combines LangGraph and CrewAI along with rule-based. In this approach, task success rate was achieved about 96.1 % and token cost reduced by 76.2%. The results also shows that decisions was faster 14.5 times than an approach where LLM is invoked for every

decision. The results analysis in our previous approach also show that approximately linear cost growth, $O(n)$, was achieved as compared to the $O(n^2)$ growth associated with the conventional multi-agent design examined in that study. The promising numbers shown in the our published approach is an evidence from the previous task setting, not assumed outcomes for 6G. We will extend our previous approach in this research and will test whether comparable benefits hold for network management.

3. **Compare fixed and adaptive thresholds.** We will compare a fixed threshold with adaptive policies based on contextual bandits, reinforcement learning, or explicitly defined network-state indicators. To evaluate the proposed solution, we will include scenarios like traffic surges, partial outages, and degraded links. One of the basic objectives we want to achieve from this research to balance between reliability, delay, and cost enough to justify its additional complexity. Based on the proposed evaluation mechanism, the reported analysis will then establish whether adaptation mechanism achieved the mentioned objectives.

4. **Test the system in disaster and mission-critical situations.** The tests will be carried out in the conditions such as failure of the mobile tower, providing temporary coverage with the help of unmanned aerial vehicles (UAVs). We will also test the system in the case of interference or disruption of tactical communication links. Evaluation will be carried out according to three criteria, i.e. the success of the task execution, decision-making time, and computational cost. We will compare the trade-offs between them. CrewAI, AutoGen, and LangChain will be evaluated separately. Each experiment will be conducted using different random seeds in order to verify its results.

5. **Test the system in two stages.** In the first stage, we will conduct experiments using a simplified multi-agent simulation and analyze the contribution of each component to the performance of the proposed approach. In the second stage, we will conduct the evaluation using ns-3 with 5G-LENA and connect the proposed escalation method to the O-RAN-based control process. These experiments will contain scenarios which are both near real-time and non real-time control functions. Open-source code and evaluation framework will be developed during the project to allow to reproduce the results of experiments by other researchers.

How we will judge success. The static hybrid model will be considered successful if it achieves superior performance compared to rule and LLM models in at least two of the three metrics under all the five tasks. With respect to adaptive thresholds, we will present their results regardless of whether their use leads to better or poorer performance. This will aid us in understanding whether there is any gain from using adaptation. The second phase will also involve comparing the ranks of policies in ns-3 to those of the first simulation. Similar rankings will support the consistency of the findings, even if the actual values differ.

Scope. This project is concerned with decision-making related to network management and choice of using either rules or LLM reasoning. In addition, the method will be evaluated in different stages of development and testing starting from the simulation and ending at the O-RAN-compliant network environment. The project will not deal with physical layer signal processing, radio hardware design or training of the foundation model. LLMs that already exist will be used via their API as done in the previous work. Comparison of the agent frameworks will not be repeated but the escalation approach will be developed and tested for the 6G networks.

Research Significance and the Challenges Addressed *(381 of 500 words)*

Why it matters. The proposed research has three principal areas of significance: operational autonomy, resource efficiency, and resilience under disruption.

First, selective escalation may strengthen operational autonomy. A network-control architecture that depends on a cloud-hosted LLM for every decision can become vulnerable to connectivity loss, service availability, or external service constraints. Keeping routine decisions on local rule-based controllers and

reserving LLM calls for cases that warrant them could reduce this dependence. The extent of that benefit will depend on the placement of the models and the availability of local execution, and will be examined as an architectural consideration rather than assumed in advance.

The second is cost and speed. Second, the work addresses the cost and timing of AI-assisted control. In the team's published evaluation, selective escalation reduced token cost by 76.2% and produced decisions 14.5 times faster than an always-LLM baseline. Network management has different workloads and service constraints, so these results should not be treated as a forecast. The project will quantify how much of the observed efficiency transfers to 6G tasks and identify the conditions under which it does not.

The third is resilience. Third, resilience is evaluated under the conditions that make network management most difficult. Disasters, partial outages, degraded connectivity, and contested links will be included in the core experimental design rather than treated as optional demonstrations. This will help determine whether selective reasoning remains dependable when network state and service priorities change rapidly.

The topic is timely given current research interest in agentic AI, autonomous network management, and AI-native 6G architectures. Publication venues will be selected according to the final contribution and the calls for papers active at the time of submission; any special-issue targets will be verified before the manuscript is prepared.

Challenges we address. A first challenge is threshold reliability. A fixed escalation threshold may route difficult cases to the inexpensive pathway, reducing cost while allowing decision quality to deteriorate without an obvious signal. We will compare fixed and adaptive thresholds and investigate calibrated confidence estimates to quantify uncertainty around the escalation decision. The aim is to make misrouting risk measurable, rather than relying solely on empirical threshold tuning.

A second challenge is methodological: choosing between a rule and an LLM is not simply a smaller version of model-to-model routing. Rules have low execution overhead and well-defined behaviour within their intended operating conditions, whereas LLM reasoning may help with context-rich or unfamiliar cases but introduces additional cost and variability. The proposed framework must therefore show measurable value against both rule-only and LLM-only baselines.

Third, simulation realism. A third challenge is evaluation realism. Abstract agent simulations are useful for rapid development, but by themselves they cannot establish how the approach behaves under network-level constraints. The staged use of ns-3 and 5G-LENA, followed by an O-RAN-aligned control-loop mapping where feasible, is intended to test whether the findings remain consistent in a more representative environment.

Fourth, real networks are not stable. Jammed links, cascading failures and sudden surges break the assumptions that make fixed thresholds work in the lab. We make these conditions a core part of the evaluation. We want to know whether the escalation method bends or breaks in exactly the conditions defence users face.

Research Methodology *(840 of 1000 words)*

The methodology is organised into two 12-month phases. Phase 1 develops and evaluates the escalation policy in a controlled multi-agent simulation; Phase 2 tests the selected design in a network simulator and maps it to an O-RAN-aligned control loop. Across both phases, experiments will use multiple random seeds, separate implementations of the selected agent frameworks, and independently generated ground truth for checking decisions. These measures are intended to make comparisons reproducible and reduce the risk of drawing conclusions from a single run or a circular evaluation procedure.

Phase 1: abstract 6G simulation (months 1 to 12). We will extend the simulation and evaluation code developed for the team's published hybrid architecture. The environment will represent five network-management tasks: admission control, slice allocation, handover, self-healing, and energy

management. Scenarios will vary in difficulty, from routine decisions with clear policy matches to cases involving uncertainty, interacting constraints, or conflicting objectives. Where suitable published 6G management benchmarks are available, their task structures will inform the scenario design. Additional scenarios will represent disaster response and tactical operating conditions, including contested links, partial outages, and competing service priorities. The assumptions and scenario-generation procedures will be documented so that other researchers can reproduce the tests.

We compare four policies.

1. **Rule only.** Deterministic controllers of the kind used in current network management.
2. **LLM only.** An agent that reasons about every decision. We build it three times, with CrewAI, AutoGen and LangChain, because our earlier work showed that the framework changes cost and delay in a major way.
3. **Static hybrid.** Our published escalation design with a fixed threshold.
4. **Adaptive hybrid.** The same design, but the threshold comes from a contextual bandit, a light reinforcement learning method, or a function of network state.

Each policy will be evaluated across multiple random seeds and three broad scenario groups: routine operation, network outages, and contested or degraded links. The primary measures will be task success, decision latency, and token or compute cost per decision. Rather than selecting a winner on a single metric, we will compare the trade-offs jointly using Pareto frontiers and report results by task and scenario type. This breakdown is important because escalation may be beneficial for some decisions yet add overhead without improving outcomes for others. The analysis will therefore report negative and mixed findings as well as improvements.

The empirical study will be complemented by a scaling and reliability analysis. We will examine how inference cost and coordination overhead change as the number of network elements or agents increases, and test whether the observed growth is closer to $O(n)$ or $O(n^2)$ under the workloads considered. Calibrated confidence methods will also be investigated to estimate the risk that a complex case is incorrectly assigned to the rule-based pathway. Any formal bound will be stated only where the assumptions and data support it; otherwise, the study will report empirical calibration and error rates rather than overstate a theoretical guarantee.

Phase 2: standards aligned validation (months 13 to 24). In Phase 2, we will implement the escalation mechanism and the strongest Phase 1 policy in ns-3 with 5G-LENA, using its 5G NR simulation capabilities. The same management decisions will then be mapped to an O-RAN-aligned control loop. The proposed design places policy configuration and threshold management in a non-real-time function, while the time-sensitive escalation decision is associated with the near-real-time control workflow, subject to the capabilities of the available implementation. The evaluation will compare the proposed approach with rule-based and reinforcement-learning baselines for tasks such as slicing and admission control. Network-level measures—including latency, throughput, service-level-agreement violation rate, and energy use—will be reported alongside inference cost. If the implementation schedule and resources permit, a UAV-assisted coverage-recovery scenario following a cell outage will be added and compared with published multi-agent RL baselines.

Tools and resources. Phase 1 will use Python and extend the existing simulation environment and evaluation harness. CrewAI, AutoGen, and LangChain will be accessed through a common LiteLLM interface where technically supported. The harness will record token usage and calculate inference cost from the applicable pricing, with the pricing source and date documented for reproducibility. Phase 2 will use the open-source ns-3 simulator with 5G-LENA. An O-RAN Software Community RIC reference implementation will be considered if access and integration requirements permit. Experiments will run on

university workstations and servers, with GPU resources used where available. LLM API expenditure will be tracked separately and managed through usage limits.

Success criteria and statistics. For each policy, we will report task success, median and 95th-percentile decision latency, and token or compute cost per decision. Aggregate estimates will include confidence intervals across random seeds, with per-task results retained to reveal differences hidden by averages. The static hybrid will meet the predefined validation criterion if it outperforms both pure baselines on at least two of the three principal measures across all five tasks. Adaptive policies will be assessed against the fixed-threshold design using the same measures. Evidence of improvement and evidence that adaptation offers no meaningful benefit will both be reported transparently.

Verification and risk. We will verify decision outcomes against ground truth generated independently of the escalation policy, reducing the possibility that evaluation labels inadvertently influence the routing decision. The protocol will also guard against conclusions based on a single random seed. If the ns-3 implementation or O-RAN integration takes longer than anticipated, the Phase 1 framework and results will remain a self-contained research output, while the Phase 2 scope will be prioritised around the most informative network scenarios. API expenditure will be controlled through batching where appropriate, call limits, and budget caps. To reduce the risk that custom scenarios are unrealistic, their structure and complexity will be compared with published 6G management benchmarks and documented for external scrutiny.

Team. The principal investigator will lead the research design, escalation-policy development, and scaling and reliability analyses. A second faculty member will lead the ns-3 implementation and O-RAN-aligned integration. Graduate and undergraduate students, working under supervision, will support scenario development, experimental runs, framework comparisons, and statistical analysis. This arrangement will also provide practical research training in agentic AI, 6G simulation, and reproducible experimental methods.

Expected Outcomes *(Word count to be checked)*

1. An open-source framework for 6G network management. It will decide when a task can be handled by rules and when an LLM is needed. We will compare it with rule-only and LLM-only approaches. The evaluation will cover task success, decision time, and computational cost. The framework will support admission control, network slicing, handover, self-healing, and energy management.

2. A comparison of fixed and adaptive escalation thresholds. We will test both approaches under changing network conditions, including traffic surges, partial outages, and disrupted links. We will also examine the risk of sending difficult cases to rule-based controllers. The results will show whether adaptive thresholds improve performance or offer little benefit.

3. A working prototype using ns-3 and 5G-LENA. We will connect the proposed method to an O-RAN-aligned control loop. This will help us check whether the results from the initial simulation also hold in a more realistic network environment.

4. An evaluation of the system under disaster and mission-critical conditions. The tests will include mobile cell outages, UAV-assisted coverage recovery where feasible, and disrupted tactical links. The findings will show where selective LLM reasoning is useful and where it may have limitations compared with using an LLM for every decision.

5. An analysis of how computational and coordination costs change as the number of network elements increases. We will examine whether the approximately linear cost growth ($O(n)$) observed in our earlier hybrid design also holds for 6G network management. We will compare it with the $O(n^2)$ growth observed in the conventional multi-agent design studied previously.

6. Research papers for relevant journals and conferences. Potential targets include IEEE Transactions on Network and Service Management and IEEE Journal on Selected Areas in Communications (JSAC). We will also consider suitable special issues on agentic AI and 6G in IEEE Network, IEEE Wireless Communications, IEEE Transactions on Network Science and Engineering (TNSE), and Digital Communications and Networks.

7. An open-source research package containing the simulation environment, escalation mechanism, and evaluation tools. Other researchers will be able to use this package to reproduce the experiments and test their own approaches.

8. Practical research training for students. They will work with agentic AI frameworks, 6G simulation tools, and network standards. They will also gain experience in experimental design, performance evaluation, and reproducible research.

9. A foundation for future field trials. The results may support collaboration with telecom and defence organisations interested in testing selective AI reasoning in real network environments.

10. A preliminary performance target for Phase 1 is to reduce computational or token cost by at least 50% and decision time by at least five times compared with the LLM-only approach. Task success should remain comparable. These targets are based on our earlier results, which showed a 76.2% reduction in token cost and 14.5 times faster decisions. However, we will test these targets rather than assume that the same improvements will occur in 6G networks.